Week 14 Homework

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In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred () proof problems are eligible for inclusion in the proof portfolio assignment.

Following Section 2.3 of *Discrete Mathematics, an Open Introduction.*Let $\{a_n\}$ be a sequence of real numbers. The k^{th} **forward difference** of this sequence are defined recursively:
The **first forward difference** Δa_n is an new sequence defined by:$\Delta a_n = a_{n+1} - a_n$.
The k**th forward difference** $\Delta^k a_n$ is $\Delta^k a_n = \Delta^{k-1} a_{n+1} - \Delta^{k-1} a_n$. So if the sequence for a_n is $1, 3, 5, 7, 9, ...$ then Δa_n is $3 - 1, 5 - 3, 7 - 5, 9 - 7, ... = 2, 2, 2, 2, 2, ...$

Exercises

Exercise

Find Δ and Δ^2 for the following sequences :

1. $2, 4, 6, 8, 10, 12, 14, ...$
2. $1, 4, 9, 16, 25, 36, 49, ...$

Exercise

Make up sequences that have:

1. 3, 3, 3, 3, … as its second differences.
2. 1, 2, 3, 4, 5, … as its third differences.
3. 1, 2, 4, 8, 16, … as its fifth differences.

Exercise

Find the closed form solution to the sequence using the using the polynomial fitting method for the sequence $0, 3, 8, 15, 24, 35, 48, 63, 80, ...$ which has the recurrence $a_n = a_{n-1} + 2n + 1$ with $a_0 = 0$.

Exercise

Find the closed form solution to the sequence using the using the polynomial fitting method for the sequence $1, 4, 10, 20, 35, 56, 84, ...$ which has the recurrence $a_n = 4a_{n-1} - 6a_{n-2} + 4a_{n-3} - a_{n-4}$ with $a_1 = 1, a_2 = 4, a_3 = 10$ and $a_4 = 20$. This also happens to the be the sequence given by summing up the triangular numbers. That is, $a_n = \sum_{i=1}^{n} T_n$.

Exercise

Calculate $\Delta\binom{n}{4}$. Use this to rewrite $\Delta\binom{n}{k}$ as a binomial coefficient.

Proofs

Proof

Prove $\Delta(\Delta^i a_n) = \Delta^{i+1} a_n$.

Proof

Let $b_n = n^3$. Prove $\Delta^2 b_n = 6n + 6$.

Proof

Let $b_n = n^k$. Prove $\Delta b_n = \sum_{i=0}^{k-1} \binom{k}{i} n^i$

Proof

Let $a_n = \alpha^n$ for $n \geq 0$ and α a real number. Prove $\Delta^k a_n = (\frac{\alpha - 1}{\alpha})^k a_n$ for all $k \geq 0$.

Proof

Let a_n and b_n be sequences. Prove $\Delta(a_n \cdot b_n) = a_{n+1} \Delta b_n + b_n \Delta a_n$.